Personal Meeting: Thermal-Noise Robustness of Sparse Refinement

October 19, 2026

1 Noise Model

1.1 Received Signal and SNR Definition

The received echo is

$$\mathbf{Y} = \mathbf{S} + \mathbf{W}, \qquad W_{mn} \overset{\text{i.i.d.}}{\sim} \mathcal{CN}(0, \sigma^2), \tag{1}$$

where $\mathbf{S}$ is the noise-free echo and $\mathbf{W}$ is receiver thermal noise (circularly symmetric complex Gaussian white noise). The SNR is defined on the noise-free CSA image $\mathbf{H}_S = \mathrm{CSA}(\mathbf{S})$ over the 800×800 scene Ω:

$$\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10} \frac{P_S}{\sigma^2}, \qquad P_S = \frac{1}{|\Omega|} \sum_{(i,j)\in\Omega} |H_{S,ij}|^2. \tag{2}$$

Since the CSA is unitary, $\mathrm{CSA}(\mathbf{W})$ is again i.i.d. $\mathcal{CN}(0, \sigma^2)$, so adding the noise to the echo or to the focused image is equivalent.

1.2 Relation to NESZ

$$\mathrm{SNR}_{\mathrm{dB}} \approx \overline{\sigma^0}_{\mathrm{dB}} - \mathrm{NESZ}_{\mathrm{dB}}, \tag{3}$$

where $\overline{\sigma^0}_{\mathrm{dB}}$ is the scene-averaged backscattering coefficient (averaged in the linear-power domain, consistent radiometric calibration).

1.3 Related Work

Gao et al. [1] model noisy TomoSAR observations as $\mathbf{G} = \mathbf{A}\boldsymbol{\Gamma} + \mathbf{E}$, with i.i.d. zero-mean Gaussian noise $\mathbf{E}$ added to the focused data, and evaluate SNR from 0 to 20 dB by Monte Carlo simulation. Our study instead adds the noise to the raw echo before CSA focusing.

2 Experimental Setup

2.1 Dataset and Tuning Protocol

All methods use the same protocol (Table 1).

Table 1: Tuning and testing protocol.

Item	Configuration
Dataset	HRSID (1,593 scenes)
Tuning set	100 fixed scenes
Test set	1,493 independent scenes
Primary objective	Maximize mAP@0.5
Tie-breaking	F1-score if mAP difference ≤ 0.002
Final tie-breaking	Prefer smaller parameters
Boundary handling	Expand the search range if the optimum lies on a boundary

2.2 Evaluation Metrics

Precision $= \mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$ and Recall $= \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$ (Table 2).

Table 2: Confusion matrix.

	Predicted Ship	Predicted Non-Ship
Actual Ship	TP	FN
Actual Non-Ship	FP	TN

3 Results

3.1 Detection Performance vs SNR

CSA shows high precision (≈ 0.98) but low recall (0.53 at ∞ to 0.14 at -5 dB): detected ships are mostly correct, but weak ships are missed (Fig. 1). The three regularizations keep recall above 0.81; $\ell_{1/2} + \mathrm{TV}$ is the highest at -5 dB (recall 0.883, mAP@0.5 0.943, Fig. 2).

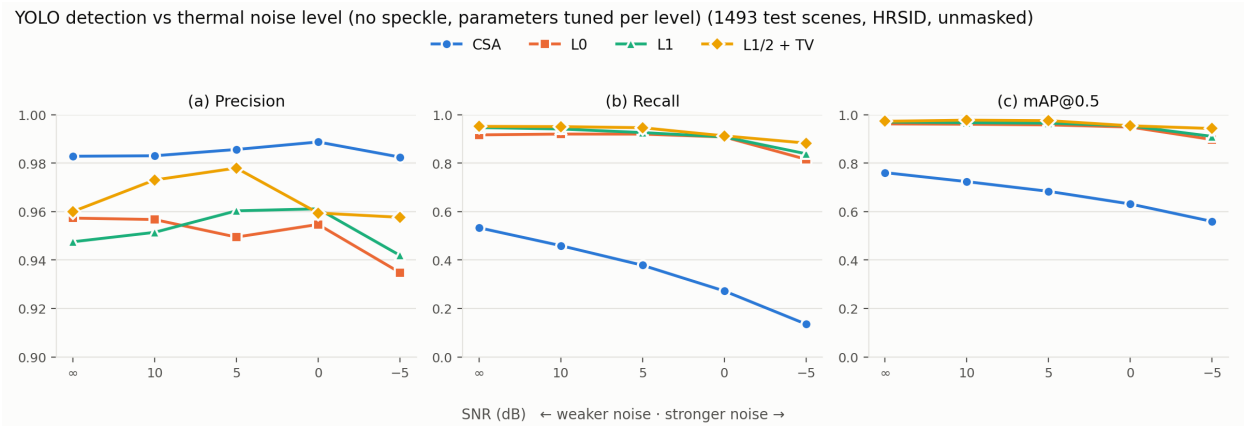


Figure 1: Precision, recall and mAP@0.5 vs SNR (parameters tuned per SNR level).

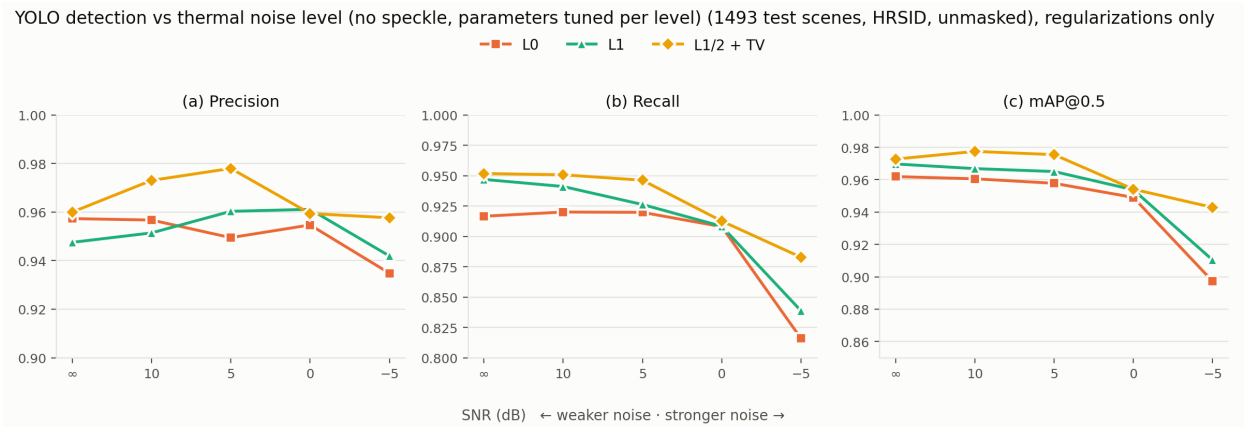


Figure 2: Same as Fig. 1, regularizations only.

3.2 Example

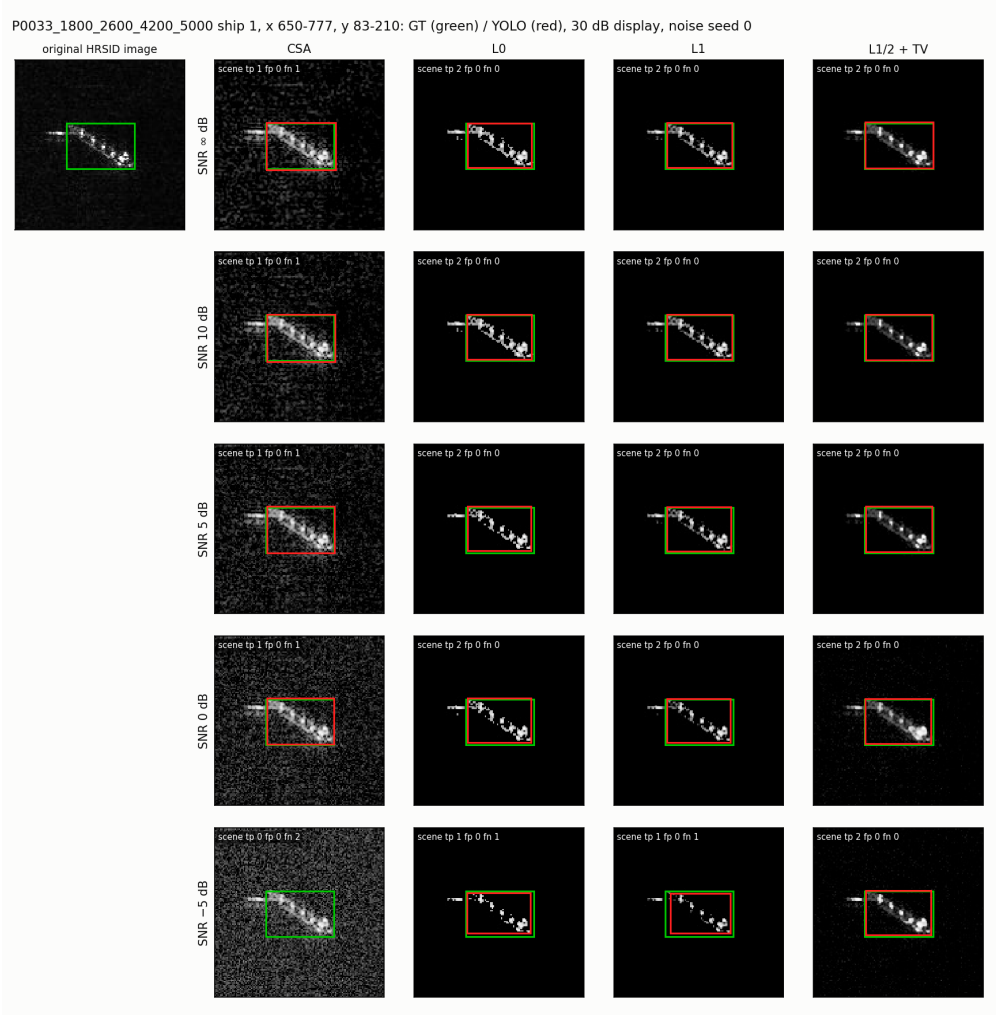


Figure 3: Scene P0033_1800_2600_4200_5000, ship 1: GT (green) and YOLO (red) boxes.

3.3 Selected Parameters

Table 3: Parameters selected per SNR level.

SNR	ℓ_0: α	ℓ_1: α	$\ell_{1/2}+\mathrm{TV}$: α	ξ	λ_{TV}
∞	0.20	0.13	0.13	2	0.03
10 dB	0.20	0.13	0.13	3	0.03
5 dB	0.25	0.16	0.13	3	0.03
0 dB	0.25	0.20	0.08	3	0.10
-5 dB	0.35	0.30	0.08	3	0.03

- ℓ_0/ℓ_1: higher noise levels require stronger thresholding (Appendix A).
- $\ell_{1/2}+\mathrm{TV}$: noise is suppressed by regularization and smoothing while weaker ship responses are retained; moderate ξ (2–3) gives stable convergence.

3.4 Parameter Robustness

Fixed: parameters tuned at ∞ and applied to all SNR levels. Adaptive: tuned per SNR level.

Table 4: Test-set mAP@0.5, fixed vs adaptive parameters.

SNR	Method	Fixed	Adaptive	Gain
0 dB	ℓ_0	0.916	0.949	+0.033
	ℓ_1	0.931	0.954	+0.023
	$\ell_{1/2}$ + TV	0.954	0.954	0.000
−5 dB	ℓ_0	0.788	0.897	+0.109
	ℓ_1	0.810	0.910	+0.100
	$\ell_{1/2}$ + TV	0.911	0.943	+0.032

$\ell_{1/2}$ + TV shows the strongest parameter robustness.

3.5 Convergence

Average iterations of $\ell_{1/2}$ + TV (tolerance 10^{-4}, cap 300; all runs converged).

Table 5: Average number of iterations of $\ell_{1/2}$ + TV.

SNR	Parameters	Average
∞	$\alpha = 0.13,\ \xi = 2,\ \lambda_{TV} = 0.03$	75
10 dB	$\alpha = 0.13,\ \xi = 3,\ \lambda_{TV} = 0.03$	93
5 dB	$\alpha = 0.13,\ \xi = 3,\ \lambda_{TV} = 0.03$	90
0 dB	$\alpha = 0.08,\ \xi = 3,\ \lambda_{TV} = 0.1$	81
−5 dB	$\alpha = 0.08,\ \xi = 3,\ \lambda_{TV} = 0.03$	56

References

[1] S. Gao, W. Wang, M. Wang, Z. Zhang, Z. Yang, X. Qiu, B. Zhang, and Y. Wu, "A robust super-resolution gridless imaging framework for UAV-borne SAR tomography," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Art. no. 5210917, pp. 1–17, 2024.

A Derivation of Soft and Hard Thresholding

This appendix derives the soft (ℓ_1) and hard (ℓ_0) thresholding operators used by the ℓ_1 and ℓ_0 methods in the experiments above.

A.1 Problem and Reduction to a Magnitude Problem

Let $\mathbf{z} = (z_1, \dots, z_N)^\mathsf{T} \in \mathbb{C}^N$ be the observed complex image and $\mathbf{x} \in \mathbb{C}^N$ the estimate. For $w \in \mathbb{C}$, $|w|$ is the modulus and $\operatorname{sgn}(w) = w/|w|$ ($\operatorname{sgn}(0) = 0$). In polar form, $x_i = \rho_i e^{j\theta_i}$ and $z_i = m_i e^{j\varphi_i}$, with $\rho_i, m_i \geq 0$. For $\lambda > 0$,

$$\hat{\mathbf{x}} = \operatorname*{arg\,min}_{\mathbf{x}\in\mathbb{C}^N} \tfrac{1}{2}\|\mathbf{x} - \mathbf{z}\|_2^2 + \lambda R(\mathbf{x}), \tag{A.1}$$

where $\|\mathbf{v}\|_2^2 = \sum_i |v_i|^2$ and $R : \mathbb{C}^N \to [0, \infty)$ is a regularizer.

Lemma 1 (Separability)**.** Assume R is separable over the entries, $R(\mathbf{x}) = \sum_{i=1}^N R_i(x_i)$ with $R_i : \mathbb{C} \to [0, \infty)$. The data term is separable as well, $\|\mathbf{x} - \mathbf{z}\|_2^2 = \sum_i |x_i - z_i|^2$, so the objective of (A.1) is $\sum_i J_i(x_i)$ with $J_i(x) = \frac{1}{2}|x - z_i|^2 + \lambda R_i(x)$, and each $\hat{x}_i = \arg\min_{x\in\mathbb{C}} J_i(x)$ is found independently.

Drop the index i. With $x = \rho e^{j\theta}$ and $z = m e^{j\varphi}$,

$$|x - z|^2 = \rho^2 + m^2 - 2\rho m \cos(\theta - \varphi). \tag{A.2}$$

Lemma 2 (Phase)**.** Assume further that each R_i depends only on the modulus, $R_i(x) = r(|x|)$ for some $r : [0, \infty) \to [0, \infty)$ with $r(0) = 0$. Then for fixed $\rho > 0$ the penalty does not depend on θ, and (A.2) is smallest at $\theta = \varphi$, where $|x - z|^2 = (\rho - m)^2$.

Combining Lemmas 1 and 2, the minimizer keeps the phase of z and thresholds its magnitude:

$$\hat{x} = \operatorname{sgn}(z)\,\hat{\rho}, \qquad \hat{\rho} = \operatorname*{arg\,min}_{\rho\geq 0}\ g(\rho), \qquad g(\rho) = \tfrac{1}{2}(\rho - m)^2 + \lambda r(\rho), \quad m = |z|. \tag{A.3}$$

A.2 ℓ_1: Soft Thresholding

$R(\mathbf{x}) = \|\mathbf{x}\|_1 = \sum_i |x_i|$ satisfies Lemmas 1 and 2 with $r(\rho) = \rho$. Then $g(\rho) = \frac{1}{2}(\rho - m)^2 + \lambda\rho$ is convex on $\rho \geq 0$, with $g'(\rho) = \rho - m + \lambda$. If $m > \lambda$, then $g' = 0$ at $\rho = m - \lambda > 0$. If $m \leq \lambda$, then $g' > 0$ for all $\rho > 0$, so the minimizer is $\rho = 0$. Hence $\hat{\rho} = \max(m - \lambda, 0)$:

$$\mathcal{S}_\lambda(z) = \operatorname{sgn}(z)\,\max(|z| - \lambda,\, 0) = \begin{cases} \big(|z| - \lambda\big)\dfrac{z}{|z|}, & |z| > \lambda, \\ 0, & |z| \leq \lambda. \end{cases} \tag{A.4}$$

Retained entries are shrunk by λ in magnitude.

A.3 ℓ_0: Hard Thresholding

$R(\mathbf{x}) = \|\mathbf{x}\|_0 = \#\{i : x_i \neq 0\}$ counts the nonzero entries. Since $x_i \neq 0 \Leftrightarrow |x_i| > 0$, it satisfies Lemmas 1 and 2 with $r(\rho) = \mathbf{1}[\rho > 0]$, and $g(\rho) = \frac{1}{2}(\rho - m)^2 + \lambda\,\mathbf{1}[\rho > 0]$. Compare the two cases: $\rho = 0$ gives $g(0) = \frac{1}{2}m^2$; for $\rho > 0$ the penalty is constant, so the best value is $\rho = m$, with $g(m) = \lambda$. $\rho = m$ is better iff $\lambda < \frac{1}{2}m^2$, i.e. $m > \sqrt{2\lambda}$:

$$\mathcal{H}_{\sqrt{2\lambda}}(z) = \begin{cases} z, & |z| > \sqrt{2\lambda}, \\ 0, & |z| \leq \sqrt{2\lambda}. \end{cases} \tag{A.5}$$

At $|z| = \sqrt{2\lambda}$, both 0 and z are minimizers; (A.5) picks 0. Retained entries are unchanged.